

Abdul Khurram

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EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo, Top 5% Internationally

2021

WORK EXPERIENCE

Shopify | *Shell, Nix, Bats, ShellCheck*

Sept. 2026 – Present

Software Engineer Intern

Toronto, ON, Canada

- Built a **Codex integration used across several teams**, handling isolated runtime environments, session-end timeouts, and reinstall behavior across Nix builds.
- Centralized shared-worktree rules-file resolution with **symlink rejection**; added shell linting and Bats tests and validated install/upgrade paths in a sandbox.

RBC Royal Bank | *Python, Java/Scala, Apache Spark, Linux*

Jan. – Aug. 2026

Software Developer Co-op

Vancouver, BC, Canada

- Designed and shipped Python/Java services that process **50M+ daily transactions** against petabyte-scale data, modeling complex entity relationships across multiple financial product lines.
- Optimized **Spark** jobs to cut **test runtime 40%**; rolled back the change after production instability.
- Architected idempotent, exactly-once batch pipelines with deterministic retry semantics and automated failover, ensuring correctness guarantees across distributed banking workloads.

Contello.ai | *Python, TypeScript/Node.js, AWS*

Jan. – Dec. 2025

Software Engineer Intern

Toronto, ON, Canada

- Designed async processing pipelines (SQS, S3) handling **thousands of jobs/day**; decoupled producers from consumers, implemented dead-letter queues, and reasoned about ordering and delivery guarantees.
- Built end-to-end test infrastructure (E2E + integration) across containerized service stacks, gating every release through CI and catching failure modes before production.
- Raised service reliability to **99.9%** by analyzing failure distributions, adding structured retry with exponential backoff, and wiring alerting on latency percentiles and error budgets.

RESEARCH EXPERIENCE

UBC Emerging Media Lab | *C++, Python, Unreal Engine*

Sept. – Dec. 2024

Software Developer & Team Lead

Vancouver, BC, Canada

- Led an interdisciplinary research team (CS × Nursing) building an AI-driven VR simulation in C++ for clinical training; scoped requirements, designed the system, and managed delivery.
- Designed a multithreaded supervisory architecture coordinating concurrent AI actors with shared state, lock-free message passing, and real-time protocol enforcement.
- Profiled the inference pipeline, identified memory allocation hotspots, and restructured the call graph to achieve **40% latency reduction** and **90% cost savings**.

PROJECTS

Backer | *TypeScript, Python, DynamoDB, XGBoost*

Hack The Coast 2026

- Modeled 10+ entity types with complex many-to-many relationships and transactional consistency constraints on a NoSQL store, designing typed repository abstractions over DynamoDB.
- Trained an XGBoost ranking model on 6 engineered features (stage affinity, text similarity, tag overlap, engagement signals) to score and surface relevant startups from event-stream data.

Workbase | *TypeScript, PostgreSQL, AWS Bedrock*

Personal Project

- Designed a structured output pipeline with LLM generation, schema validation, automated repair, evidence attribution, deterministic retry, and audit logging.
- Wrote test suites covering domain state machines (status transitions, eligibility rules, deduplication logic) and the structured LLM client (parsing, repair, validation).

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Scala, TypeScript/JavaScript, SQL

Tools & Platforms: Apache Spark, PostgreSQL, AWS (Lambda, DynamoDB, S3, SQS), Docker, Git, Linux